

Database Management Systems CPS 3740

Lecture 0

Computer Science, Kean University

Dr. Ching-yu (Austin) Huang

Instructor Paolien Wang

Important notes

- This slide set is for self-study.
- The instructor will quickly go through some slides.
- Students must follow the slides and do the hands-on exercises.
- The project URL must be completed in the first week.
- Please see class notes posted on OneDrive.

Software Tools for assignments

- Below tools are required for project & homework
 - Tools connect to Linux web server (obi2.kean.edu)
 - Windows User: PuTTY
 - Mac Users: Terminal
 - Tools connect to MySQL (database server: imc.kean.edu)
 - Before midterm, using MySQL command-line on obi2 server
 - After midterm, using a MySQL WorkBench GUI tool
 - Tools to transfer files between Windows and Linux
 - FileZilla
 - Tools to edit HTML and PHP
 - Sublime
- Refer to the documents under “tools” folder on the class shared drive

obi2.kean.edu server

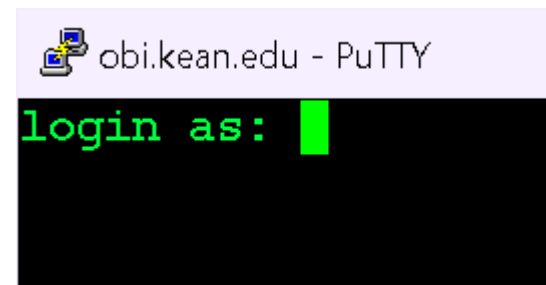
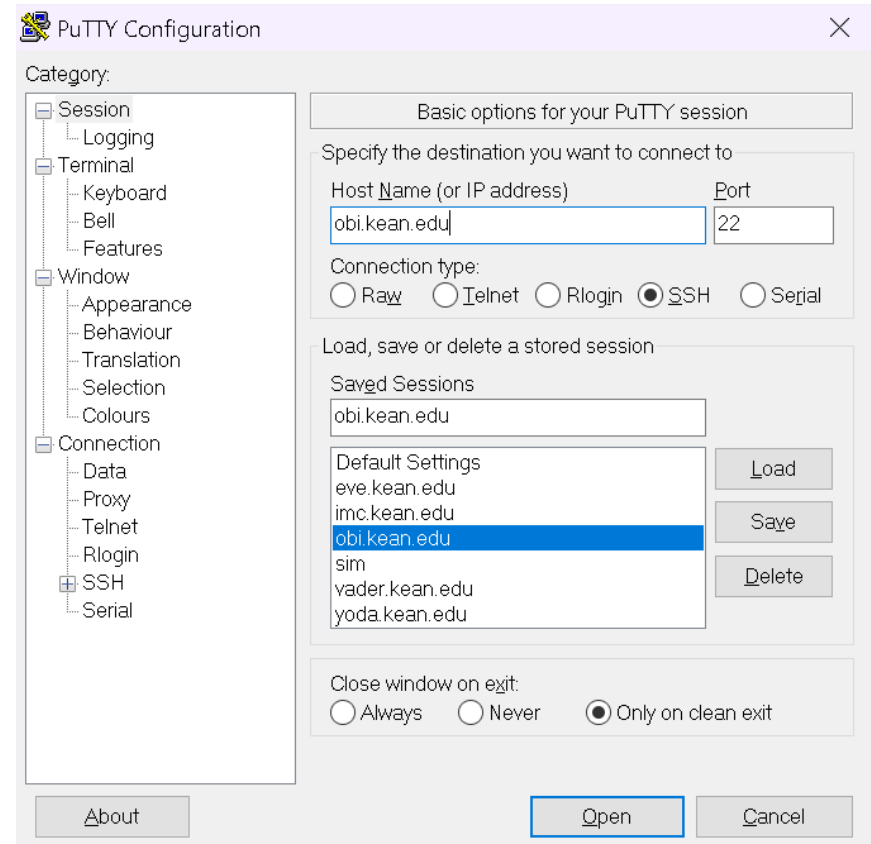
- **obi2.kean.edu** (alias **obi.kean.edu**) is a Linux system that hosts a **web server** for our assignments and projects and provides a **MySQL client** for connecting to our database server via the command line.
 - IP: **192.168.122.1** (private network)
 - **Must use Kean VPN to access if you try to login from home.**
 - Also called it the Application server
 - Your project will be implemented on it.
- Login: **your Kean email with @kean.edu**
- Password: **your Kean email password**
- We will do SQL command exercises on it.
 - You need to login to the system
- If you have trouble to login to your obi account, please let your instructor know ASAP.

Tools connect to Linux

- You will need a console to connect and work on a Linux server.
- **Terminal** is for **Mac** computer.
- **PuTTY** is for **Windows** computer
 - using SSH protocol for Windows and Unix platforms, along with an xterm terminal emulator.
 - <https://www.putty.org/>

Tool connect to and login to obi2 server #1 (Windows User – PuTTY)

1. Use Putty to connect to **obi2.kean.edu**
– type in Hostname (**obi2.kean.edu**), port (22)
2. You will be asked to input the Account login (your kean email ID) and password. Type it in and press Enter.
NOTE: The cursor will NOT be moving when you are entering your password.



Tool connect to and login to obi2 server #1 (Windows User – PuTTY)

3. If the SSH connection to obi.kean.edu has been successfully established, you will see “Last login: <timestamp> ...” and a Linux command-line prompt \$.

Login: your Kean email with @kean.edu

Password: your kean email password

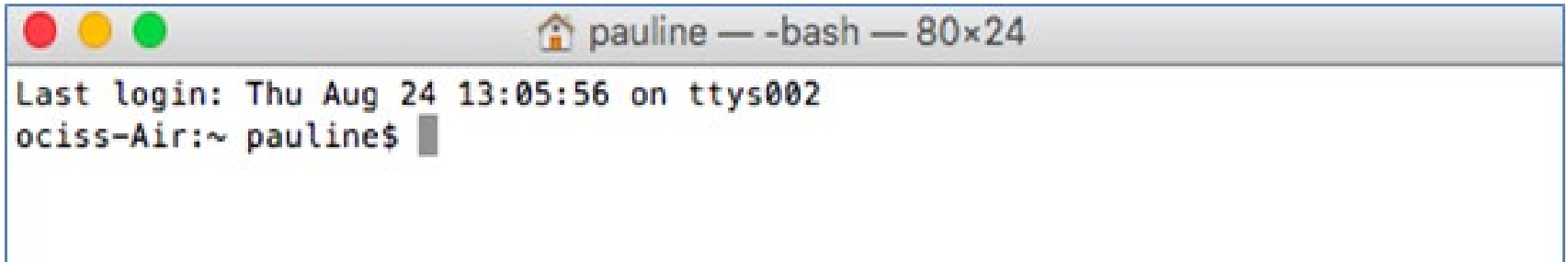
Tool connect to and login to obi2 server #1 (Mac User – Terminal)

- If you use Mac OS X, you don't need to install a third-party client like PuTTY to connect to your hosting account via Secure Shell (SSH). Instead, you can use Terminal - a command line emulation program.
- Before you start, make sure [SSH access is enabled](#) for your hosting account.
 1. Open your Terminal application: (You can enter terminal in the search)



- You will see a window with a **\$** sign and a blinking cursor. This is your basic command prompt. You can establish an SSH connection to your hosting server:

Tool connect to and login to obi2 server #2 (Mac User – Terminal)

A screenshot of a Mac Terminal window. The title bar shows a home icon, the name 'pauline', and the command '-bash' with a window size of '80x24'. The terminal content shows 'Last login: Thu Aug 24 13:05:56 on ttys002' followed by the prompt 'ociss-Air:~ pauline\$' with a cursor.

1. To connect to **obi2.kean.edu**, you need to execute the following command after the prompt \$

ssh xxxx@kean.edu@obi2.kean.edu

PS: xxxx@kean.edu is your **Kean email**

2. Now the computer will attempt to connect to your hosting server. If it is your first time connecting to the server, you will see a message asking you to confirm the connection establishment. Type **yes** and press the Enter key to proceed.

Tool connect to and login to obi2 server #3 (Mac User – Terminal)

3. You will be asked to input the SSH login and password.
Type it in and press Enter.

NOTE: The cursor will **NOT** be moving when you will be entering your password.

4. If all is good, you will see a welcome message and a line beginning with `[xxxx@obi ~]$` , this will indicate that the SSH connection to obi.kean.edu has been successfully established.

Now, you successfully login to obi.kean.edu and you have to use Unix commands after the prompt `$`.

Project & tools

(Day 03 Project Discussion Reference)

- Project (*.php, *.html) should be on
 - Webserver host: obi2.kean.edu
 - SSH port#: 22
- Tools to connect to the web server
 - Windows: FileZilla, PuTTY
 - Mac: Terminal and FileZilla
- Database should be based on
 - Database host: imc.kean.edu
 - Mysql port#: 3306
- Both login user ID: your Kean email ID
- Password: your Kean ID

Create project website folders in server

- Follow previous slides to connect to the application server
- Type the following under Linux command line prompt \$ to create folder to place your code

1. Go to the home directory, and change the permission to 705 to home directory

\$ cd

\$ pwd → show current path, make sure it's your home directory

\$ chmod 705 . → do not miss the dot, . (dot) means your current directory

→ Important: Always ensure the owner's permission is set to 7 so you can read, write, execute (chmod 7xx <dir/file>)

2. Make public_html folder, go to public_html, and change the permission to 705

\$ cd → return to your home directory, make sure you are in home directory now

\$ mkdir public_html → create public_html folder under your home directory, pay attention that linux system is case sensitive

\$ cd public_html

\$ chmod 705 .

3. Make CPS3740 folder under public_html, go to CPS3740, change the permission to 705

\$ mkdir CPS3740

\$ cd CPS3740

\$ chmod 705 .

4. go to each folder (your home directory, public_html, and CPS3740), verify your folder permission is open to 705, see the screen shot below for the example to check public_html

\$ ls -la

```
pawang@kean.edu@obi2 ~]$ cd public_html
pawang@kean.edu@obi2 public_html]$ ls -la
total 24
drwx---r-x+ 6 pawang@kean.edu 481800513   76 Sep 13 14:52 .
drwx---r-x+ 7 pawang@kean.edu 481800513 4096 Sep 14 00:47 ..
```

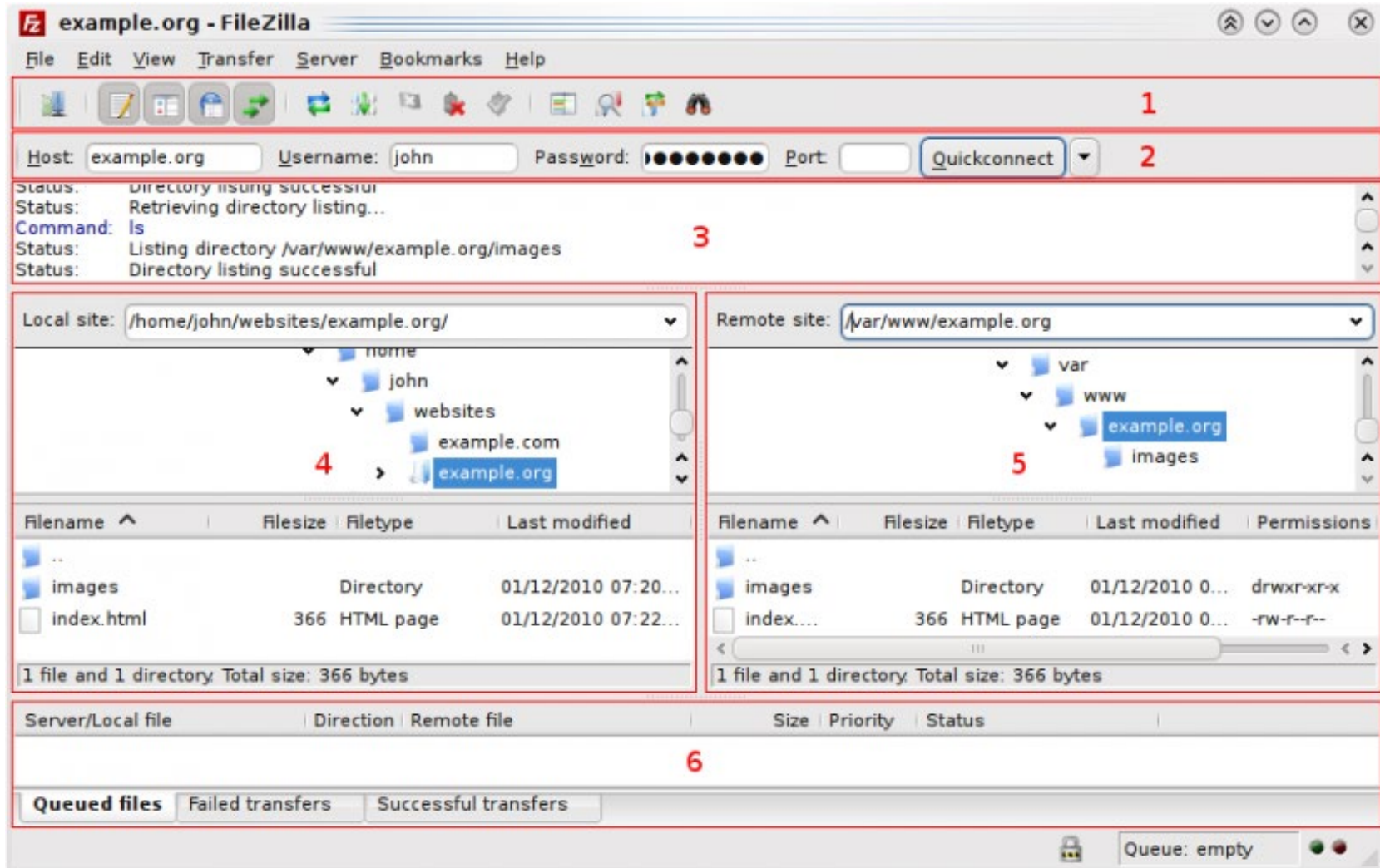
← 705

← current directory

Transfer files to a Linux System (FileZilla)

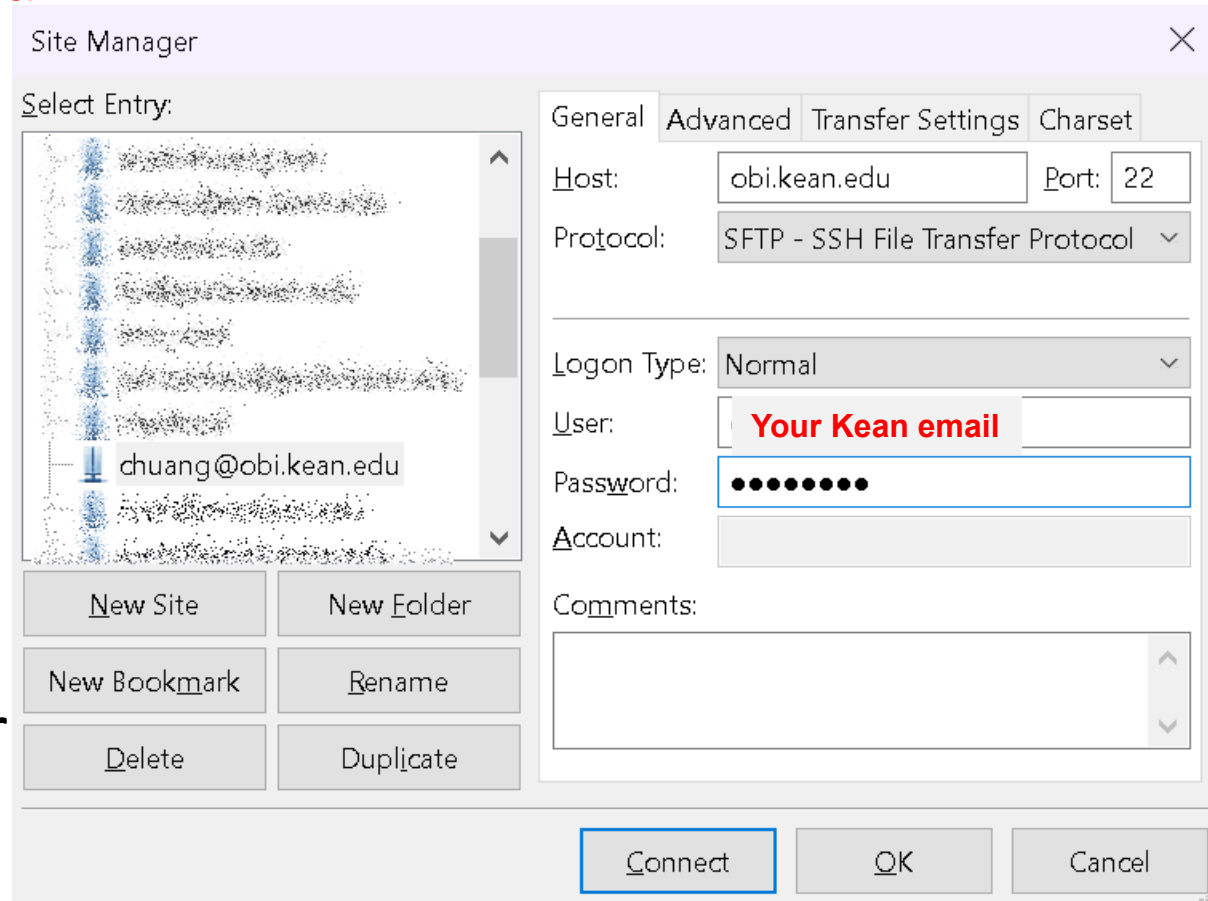
- Download, install and use FileZilla software
 - For both Windows and Mac
- Create folder public_html under your home
- Create folder CPS3740 under public_html
- Put the HTML and PHP files under CPS3740 folder
- On the right panel (Linux side), right click on the mouse to change permission mode
 - Folders: your home, public_html, CPS3740
 - Files: *.html, *.PHP

- Tutorial: [https://wiki.filezilla-project.org/FileZilla_Client_Tutorial_\(en\)](https://wiki.filezilla-project.org/FileZilla_Client_Tutorial_(en))
- Legend: 1. Toolbar, 2. Quick connect bar, 3. Message log, 4. Local pane, 5. Remote pane, 6. Transfer queue



FileZilla – create a new site

- File menu → Site Manager → New Site
- Host: **obi2.kean.edu**
- Port: 22
- SSH protocol
- Normal type
- Enter your login
- Enter password
- Click “OK”
- Go to site manager
- Click on the entry



FileZilla: Transfer Files

- Use sublime to edit the PHP and HTML files
 - Use FileZilla to transfer files between Windows and Linux
 - All html and php files should be under `public_html/CPS3740`
- Drag and drop to transfer files
 - Double check the source and destination - overwrite
- On Linux
 - HTML, PHP file permission mode should be `704`
 - PHP can be `705`

Exercise: Your project URL – part 1

- Create an index.html file using sublime

```
<HTML>
Welcome to your_name CPS 3740 project.
</HTML>
```
- Use FileZilla to transfer the index.html to your CPS3740 folder
 - Place all your project files (*.php, .html*) in the **CPS3740** folder. Do NOT create subfolders within CPS3740 to store your files.

Exercise: Your project URL – part 2

- On the Unix terminal, type the following
 - Go to your home directory
`$ cd`
 - Change your home directory permission mode. Do NOT miss the dot. This step must be done on the Linux terminal, NOT using FileZilla.
`$ chmod 705 .`
 - change permission mode recursively
`$ chmod -R 705 public_html`

P.S.: You can refer to the slide provided in this PowerPoint for a step-by-step example -- **'Creating project website folders on the server.'**

Exercise: Your project URL – part 3

- Use browser to open.
 - You must have index.html inside the folder.
Otherwise, there will be a forbidden or permission denied error

<https://obi.kean.edu/~xxxx@kean.edu/CPS3740/>

- xxxx@kean.edu is your Kean email
- If your project URL is not working, please repeat parts 1 and 2.
- You must make your project URL working in the first week.

Text Editor

- You need a plain text editor to edit the PHP, SQL, HTML files.
 - Do NOT use MS-office: **special** characters and **single** quote and **double** quote problem.
- Vim on Unix
 - Tutorial: <http://www.openvim.com/tutorial.html>
 - <https://linuxconfig.org/vim-tutorial>
- Sublime on Windows, Mac
 - <https://www.sublimetext.com/download>
 - You need to manually transfer files to the web server

Filezilla

- A cross-platform graphical FTP, SFTP, and FTPS file management tool for Windows, Linux, Mac OS X, and more.
- Use the client software to transfer files between Windows/Mac (**your PC**) and Unix (**instructor's web server**).
- Download at

<https://filezilla-project.org/>